Assignment set I

Thursday, 14 May 2026 14:20

1.1) $\varphi(\lambda) = \det(A - \lambda I) = 0$ charactaristic polinomial

$A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$

$\varphi(\lambda) = \left| \begin{pmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{pmatrix} \right| = (2-\lambda)(4-\lambda) - 3 \cdot 1 = \lambda^2 - 6\lambda + 5 = 0$

$\lambda_1 = 5, \quad \lambda_2 = 1$

$B = \begin{pmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{pmatrix}$

$\varphi(\lambda) = \left| \begin{pmatrix} \cos(\alpha) - \lambda & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) - \lambda \end{pmatrix} \right| = (\cos(\alpha) - \lambda)^2 + \sin(\alpha)^2 = \lambda^2 - 2\lambda\cos(\alpha) + \overbrace{\cos(\alpha)^2 + \sin(\alpha)^2}^{1} =$

$= \lambda^2 - 2\lambda\cos(\alpha) + 1$

$\lambda_{1,2} = \frac{2\cos(\alpha) \pm \sqrt{4\cos^2(\alpha) - 4}}{2} = \cos(\alpha) \pm \sqrt{\cos^2(\alpha) - 1}$

$C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}$

$\varphi(\lambda) = \left| \begin{pmatrix} 1-\lambda & 2 & 3 \\ 2 & 1-\lambda & 3 \\ 3 & 3 & 6-\lambda \end{pmatrix} \right| = (1-\lambda)^2(6-\lambda) + \overset{18}{2 \cdot 3 \cdot 3} + \overset{18}{3 \cdot 2 \cdot 3} - \overset{9}{3 \cdot 3 (1-\lambda)} - \overset{9}{3 \cdot 3 \cdot (1-\lambda)} - \overset{4}{2^2 (6-\lambda)} =$

$= (1 - 2\lambda + \lambda^2)(6-\lambda) + 36 - 18(1-\lambda) - 4(6-\lambda) = 6 - 12\lambda + 6\lambda^2 - \lambda + 2\lambda^2 - \lambda^3 + 36 - 18 + 18\lambda - 24 + 4\lambda =$

$= -\lambda^3 + 9\lambda^2 + 9\lambda = \lambda(-\lambda^2 + 9\lambda^2 + 9) = 0$

$\lambda_1 = 0 \qquad \lambda_{1,2} = \frac{-9 \pm \sqrt{64 + 36}}{-2} = 4 \pm 5$

$\lambda_2 = 9, \quad \lambda_3 = -1$

1.2) $m\ddot{x} + b\dot{x} + kx = 0 \qquad m > 0, b \geq 0, k > 0$

a) lets introduce $v = \dot{x}$, so $\ddot{x} = \dot{v}$

$\dot{v} = \ddot{x} = \frac{-b\dot{x} - kx}{m} = \frac{-bv - kx}{m} = -\frac{b}{m}v - \frac{k}{m}x$

$$\begin{cases} \dot{x} = v \\ \dot{v} = -\frac{h}{m} v - \frac{k}{m} x \end{cases}$$

b) $A = \begin{pmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{h}{m} \end{pmatrix}$

$$\varphi(\lambda) = \left| \begin{pmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{h}{m} - \lambda \end{pmatrix} \right| = \lambda^2 + \frac{h}{m}\lambda + \frac{k}{m} = 0$$

$$\lambda = \frac{-\frac{h}{m} \pm \sqrt{\frac{h^2}{m^2} - \frac{4k}{m}}}{2}$$

so the sign of $\frac{h^2}{m^2} - \frac{4k}{m}$ determines the type of eigen value.

$\frac{h^2}{m^2} - \frac{4k}{m} < 0$	$\frac{h^2}{m^2} - \frac{4k}{m} = 0$	$\frac{h^2}{m^2} - \frac{4k}{m} > 0$
$0 \le h < 2\sqrt{km}$	$h = 2\sqrt{mk}$	$h > 2\sqrt{mk}$
$\Downarrow$		
2 complex λ	1 real λ	2 real λ

split by cases

$\underline{h = 0}$: $\lambda_{1,2} = \frac{0 \pm \sqrt{0 - \frac{4k}{m}}}{2} = \pm\sqrt{-\frac{k}{m}} = \pm i\sqrt{\frac{k}{m}}$

$$\begin{pmatrix} -i\sqrt{\frac{k}{m}} & 1 \\ -\frac{k}{m} & -\frac{0}{m} - i\sqrt{\frac{k}{m}} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = 0 \Rightarrow \begin{matrix} -i\sqrt{\frac{k}{m}} \cdot w_1 + w_2 = 0 \Rightarrow w_2 = i\sqrt{\frac{k}{m}} \cdot w_1 \\ -\frac{k}{m} w_1 - i\sqrt{\frac{k}{m}} w_2 = 0 \end{matrix}$$

$$\Downarrow$$

$$-\frac{k}{m} w_1 - i\sqrt{\frac{k}{m}} \cdot i\sqrt{\frac{k}{m}} \cdot w_1 = -\frac{k}{m} w_1 + \frac{k}{m} = 0$$

$$\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = \begin{pmatrix} 1 \\ i\sqrt{\frac{k}{m}} \end{pmatrix}$$

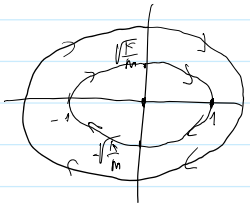
$$e^{i\sqrt{\frac{k}{m}}t} \begin{pmatrix} 1 \\ i\sqrt{\frac{k}{m}} \end{pmatrix} = \left(\cos\left(\sqrt{\tfrac{k}{m}}t\right) + i\sin\left(\sqrt{\tfrac{k}{m}}t\right)\right) \begin{pmatrix} 1 \\ i\sqrt{\frac{k}{m}} \end{pmatrix} = \begin{pmatrix} \cos\left(\sqrt{\frac{k}{m}}t\right) + i\sin\left(\sqrt{\frac{k}{m}}t\right) \\ i\sqrt{\frac{k}{m}}\cos\left(\sqrt{\frac{k}{m}}t\right) - \sqrt{\frac{k}{m}}\sin\left(\sqrt{\frac{k}{m}}t\right) \end{pmatrix}$$

real - $\begin{pmatrix} \cos\left(\sqrt{\frac{k}{m}}t\right) \\ -\sqrt{\frac{k}{m}}\sin\left(\sqrt{\frac{k}{m}}t\right) \end{pmatrix}$ imaginary - $\begin{pmatrix} \sin\left(\sqrt{\frac{k}{m}}t\right) \\ \sqrt{\frac{k}{m}}\cos\left(\sqrt{\frac{k}{m}}t\right) \end{pmatrix}$

$$x(t) = c_1 \cos\left(\sqrt{\tfrac{k}{m}}t\right) + c_2 \sin\left(\sqrt{\tfrac{k}{m}}t\right)$$

$$v(t) = -c_1\sqrt{\tfrac{k}{m}}\sin\left(\sqrt{\tfrac{k}{m}}t\right) + c_2\sqrt{\tfrac{k}{m}}\cos\left(\sqrt{\tfrac{k}{m}}t\right)$$

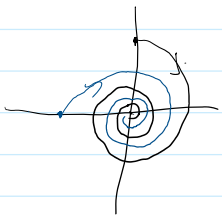
looking at $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $\quad x(t) = \cos\left(\sqrt{\frac{k}{m}}t\right)$, $v(t) = -\sqrt{\frac{k}{m}}\sin\left(\sqrt{\frac{k}{m}}t\right)$



stable trajectory

$\underline{0 > h > 2\sqrt{km}}$: $\frac{h^2}{m^2} - \frac{4k}{m} < 0 \Rightarrow \lambda_{1,2} = \frac{-\frac{h}{m} \pm i\sqrt{\frac{4k}{m} - \frac{h^2}{m^2}}}{2} = -\frac{h}{2m} \pm i\sqrt{\frac{k}{m} - \frac{h^2}{4m^2}}$

$\lambda = \alpha \pm i\beta$, $\alpha = -\frac{b}{2m}$, $\beta = \sqrt{\frac{k}{m} - \frac{b^2}{4m^2}}$



real solutions will be of shape $C_1 e^{\alpha t}\cos(\beta t) + C_2 e^{\alpha t}\sin(\beta t)$

since $b > 0$ and $m > 0$, $\alpha = -\frac{b}{2m} < 0$, so $t \to \infty \Rightarrow e^{\alpha t} \to 0$
so it spirals inwards

$b = 2\sqrt{mk}$: $\frac{b^2}{m^2} - \frac{4k}{m} = 0 \Rightarrow \lambda = -\frac{2\sqrt{mk}}{2m} \pm 0 = -\sqrt{\frac{k}{m}}$

$\begin{pmatrix} \sqrt{\frac{k}{m}} & 1 \\ -\frac{k}{m} & -\sqrt{\frac{k}{m}} \end{pmatrix}\begin{pmatrix} w_1 \\ w_2 \end{pmatrix} = 0 \Rightarrow \begin{matrix} \sqrt{\frac{k}{m}} w_1 + w_2 = 0 \Rightarrow w_2 = -\sqrt{\frac{k}{m}} w_1 \\ -\frac{k}{m} w_1 - \sqrt{\frac{k}{m}} w_2 = 0 \quad -\frac{k}{m} w_1 - \sqrt{\frac{k}{m}} \cdot -\sqrt{\frac{k}{m}} w_1 = 0 \end{matrix}$

$V_1 = \begin{pmatrix} 1 \\ -\sqrt{\frac{k}{m}} \end{pmatrix}$

we'll look for the generalized eigenvector $V_2 = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$: $(A - \lambda I)V_2 = V_1$

$\begin{pmatrix} \sqrt{\frac{k}{m}} & 1 \\ -\frac{k}{m} & -\sqrt{\frac{k}{m}} \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -\sqrt{\frac{k}{m}} \end{pmatrix}$ $\begin{matrix} \sqrt{\frac{k}{m}} u_1 + u_2 = 1 \Rightarrow u_2 = 1 - \sqrt{\frac{k}{m}} u_1 \\ -\frac{k}{m} u_1 - \sqrt{\frac{k}{m}} u_2 = -\sqrt{\frac{k}{m}} \end{matrix}$

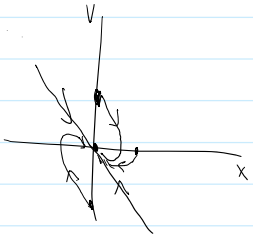
$-\frac{k}{m} u_1 - \sqrt{\frac{k}{m}} + \frac{k}{m} u_1 = -\sqrt{\frac{k}{m}}$

$V_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$x(t) = C_1 e^{-\sqrt{\frac{k}{m}}t} \cdot 1 + C_2 e^{-\sqrt{\frac{k}{m}}t}(0 + t) = C_1 e^{-\sqrt{\frac{k}{m}}t} + C_2 t e^{-\sqrt{\frac{k}{m}}t} = e^{-\sqrt{\frac{k}{m}}t}(C_1 + C_2 t)$

$v(t) = C_1 e^{-\sqrt{\frac{k}{m}}t} \cdot -\sqrt{\frac{k}{m}} + C_2 e^{-\sqrt{\frac{k}{m}}t}\left(1 - t\sqrt{\frac{k}{m}}\right) = e^{-\sqrt{\frac{k}{m}}t}\left(-C_1\sqrt{\frac{k}{m}} + C_2 - C_2 t\sqrt{\frac{k}{m}}\right)$

$t \to 0 \Rightarrow e^{-\sqrt{\frac{k}{m}}t} \to 0$, so everything still decays
$te^{-\sqrt{\frac{k}{m}}t} \to 0$



$b > 2\sqrt{mk}$: $\lambda_{1,2} = -\frac{b \pm \sqrt{b^2 - 4mk}}{2m}$

since $b > 2\sqrt{mk}$, $b^2 - 4mk$ is positive, so there are 2 real solutions
λ sign depends on $-b \pm \sqrt{b^2 - 4mk}$
$0 > -4mk$
$b^2 > b^2 - 4mk$
$b > \sqrt{b^2 - 4mk} \Rightarrow$

$\lambda_1 = \frac{-b + \sqrt{b^2 - 4mk}}{2m} < 0$

$\lambda_2 = \frac{-b - \sqrt{b^2 - 4mk}}{2m} < 0$

$\begin{pmatrix} -\lambda & 1 \\ -\frac{k}{m} & -\frac{b}{m} - \lambda \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\begin{matrix} -\lambda u_1 + u_2 = 0 \Rightarrow u_2 = \lambda u_1 \\ -\frac{k}{m} u_1 - \frac{b}{m} u_2 - \lambda u_2 = 0 \end{matrix}$

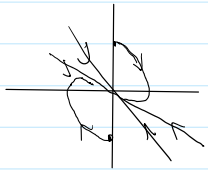
so eigenvectors are of shape - $V = \begin{pmatrix} u_1 \\ \lambda u_1 \end{pmatrix}$
let $u_1 = 1$

So eigenvectors are of shape - $V = \begin{pmatrix} u_1 \\ \lambda u_1 \end{pmatrix}$

take $u_1 = 1$

$V_1 = \begin{pmatrix} 1 \\ \frac{-b+\sqrt{b^2-4mk}}{2m} \end{pmatrix}$ $\quad V_2 = \begin{pmatrix} 1 \\ \frac{-b-\sqrt{b^2-4mk}}{2m} \end{pmatrix}$

$\begin{pmatrix} x(t) \\ v(t) \end{pmatrix} = C_1 e^{\lambda_1 t}\begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} + C_2 e^{\lambda_2 t}\begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}$



since $\lambda_1, \lambda_2 < 0$, $t \to \infty \Rightarrow e^{\lambda t} \to 0$

so the system decays to 0 here as well

1.3) $\dot{X} = AX \qquad A = \begin{pmatrix} 0 & 6 & 6 \\ 5 & 11 & -1 \\ 1 & -5 & 7 \end{pmatrix}$

it has $\lambda = 0$ if $\det(A) = 0$

$$\left|\begin{pmatrix} 6 & 6 & 6 \\ 5 & 11 & -1 \\ 1 & -5 & 7 \end{pmatrix}\right| = 6\cdot 11\cdot 7 - 6\cdot 1\cdot 1 - 6\cdot 5\cdot 5 - 6\cdot 11\cdot 1 - 1\cdot 5\cdot 6 - 7\cdot 6\cdot 5 = 462 - 6 - 150 - 66 - 30 - 210 = 0$$

So $\lambda = 0$ and there is eigenvector V s.t. $AV = 0$

when $X = cV$, $c \in \mathbb{R}$

$\dot{X} = AX = cAV = 0$, so $\dot{X}$ is 0 for any point on V. there are infinite fixed points

1.4) $A = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} \qquad \left|\begin{pmatrix} 6-\lambda & -3 \\ 2 & 1-\lambda \end{pmatrix}\right| = 6 - 7\lambda + \lambda^2 + 6 = \lambda^2 - 7\lambda + 12 = 0$

$\lambda_{1,2} = \frac{7 \pm \sqrt{49-48}}{2} = \frac{7 \pm 1}{2} \qquad \lambda_1 = 3, \ \lambda_2 = 4$

λ_1 $\begin{pmatrix} 6-3 & -3 \\ 2 & 1-3 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\quad \begin{cases} 3u_1 - 3u_2 = 0 \\ 2u_1 - 2u_2 = 0 \end{cases}$ $\quad u_1 = u_2$

$V_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $X_1(t) = e^{3t}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

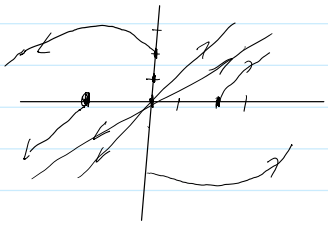
λ_2 $\begin{pmatrix} 6-4 & -3 \\ 2 & 1-4 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\quad \begin{cases} 2u_1 - 3u_2 = 0 \\ 2u_1 - 3u_2 = 0 \end{cases}$ $\quad 2u_1 = 3u_2$

$V_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $X_2(t) = e^{4t}\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

$$X(t) = c_1 e^{3t}\begin{pmatrix}1\\1\end{pmatrix} + c_2 e^{4t}\begin{pmatrix}3\\2\end{pmatrix}$$

to show the solutions are linearly independent, we will show v_1, v_2 are

$$\left|\begin{pmatrix}1 & 3\\ 1 & 2\end{pmatrix}\right| = 2-3 = -1 \neq 0$$



since e^{4t} grows faster than e^{3t}, trajectories become tangent to its eigenvector

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$$A = \begin{pmatrix}-2 & 1\\ -4 & 3\end{pmatrix} \qquad \left|\begin{pmatrix}-2-\lambda & 1\\ -4 & 3-\lambda\end{pmatrix}\right| = -6-\lambda+\lambda^2+4 = \lambda^2-\lambda-2$$

$$\lambda_{1,2} = \frac{1 \pm \sqrt{1+8}}{2} \qquad \lambda_1 = 2,\ \lambda_2 = -1$$

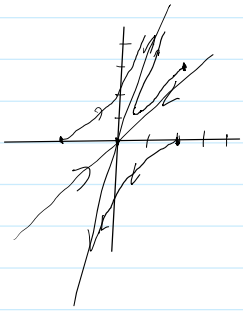
(λ_1) $\begin{pmatrix}-2-2 & 1\\ -4 & 3-2\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$ $\quad -4u_1 + u_2 = 0 \quad u_2 = 4u_1$

$$v_1 = \begin{pmatrix}1\\4\end{pmatrix}$$

(λ_2) $\begin{pmatrix}-2+1 & 1\\ -4 & 3+1\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}$ $\quad -u_1+u_2 = 0 \quad u_1 = u_2$
$\quad -4u_1+4u_2 = 0$

$$v_2 = \begin{pmatrix}1\\1\end{pmatrix}$$

$$X(t) = c_1 e^{2t}\begin{pmatrix}1\\4\end{pmatrix} + c_2 e^{-t}\begin{pmatrix}1\\1\end{pmatrix}$$



$$\left|\begin{pmatrix}1 & 1\\ 4 & 1\end{pmatrix}\right| = 1-4 = -3 \neq 0 \Rightarrow \text{linear independence}$$

$$A = \begin{pmatrix}-4 & 1\\ -1 & -2\end{pmatrix} \qquad \left|\begin{pmatrix}-4-\lambda & 1\\ -1 & -2-\lambda\end{pmatrix}\right| = 8+6\lambda+\lambda^2+1 = \lambda^2+6\lambda+9 = 0$$

$$\lambda_{1,2} = \frac{-6 \pm \sqrt{36-36}}{2} \qquad \lambda_1 = -3$$

$$\begin{pmatrix}-4+3 & 1\\ -1 & -2+3\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} \qquad -u_1+u_2 = 0 \quad u_1 = u_2$$
$$-u_1+u_2 = 0$$

$$v_1 = \begin{pmatrix}1\\1\end{pmatrix}$$

we need to find the generalized eigenvector w, $(A+\lambda I)w = v_1$

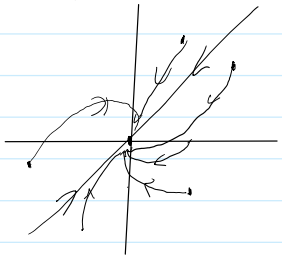
$$\begin{pmatrix}-4+3 & 1\\ -1 & -2+3\end{pmatrix}\begin{pmatrix}w_1\\w_2\end{pmatrix} = \begin{pmatrix}1\\1\end{pmatrix} \qquad -w_1+w_2 = 1 \quad w_2 = w_1+1$$
$$-w_1+w_2 = 1$$

$$\begin{array}{ccc} -4 & 1 & 1 \\ -1 & -2 & 0.9 \end{array} \qquad -3.1$$

$$w = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$X(t) = c_1 e^{3t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{3t}\left[t\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right] = c_1 e^{3t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{3t}\begin{pmatrix} t \\ t+1 \end{pmatrix}$$

$$\left|\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\right| = 1 - 0 = 1 \neq 0 \Rightarrow \text{linear independence}$$



$$A = \begin{pmatrix} 1 & -2 \\ 3 & -4 \end{pmatrix} \qquad \left|\begin{pmatrix} 1-\lambda & -2 \\ 3 & -4-\lambda \end{pmatrix}\right| = -4 + 3\lambda + \lambda^2 + 6 = \lambda^2 + 3\lambda + 2$$

$$\lambda_{1,2} = \frac{-3 \pm \sqrt{9-8}}{2} \qquad \lambda_1 = -1 \qquad \lambda_2 = -2$$

(λ_1) $\begin{pmatrix} 1+1 & -2 \\ 3 & -4+1 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$ $\quad \begin{cases} 2u_1 - 2u_2 = 0 \\ 3u_1 - 3u_2 = 0 \end{cases}$ $\quad u_1 = u_2$

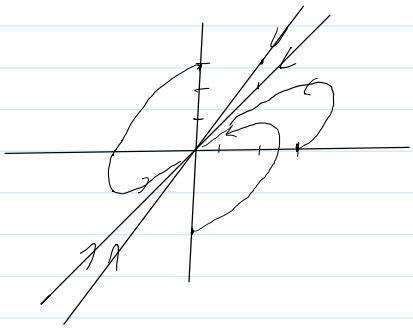
$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

(λ_2) $\begin{pmatrix} 1+2 & -2 \\ 3 & -4+2 \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = 0$ $\quad \begin{cases} 3u_1 - 2u_2 = 0 \\ 3u_1 - 2u_2 = 0 \end{cases}$ $\quad 2u_2 = 3u_1$

$$v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$X(t) = c_1 e^{-t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 e^{-2t}\begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

$$\left|\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}\right| = 3 - 2 = 1 \neq 0 \Rightarrow \text{linearly independant}$$



$$A = \begin{pmatrix} 5 & -5 \\ 2 & -1 \end{pmatrix} \qquad \left|\begin{pmatrix} 5-\lambda & -5 \\ 2 & -1-\lambda \end{pmatrix}\right| = -5 - 4\lambda + \lambda^2 + 10 = \lambda^2 - 4\lambda + 5$$

$$\lambda_{1,2} = \frac{4 \pm \sqrt{16-20}}{2} \qquad \lambda_1 = 2+i \qquad \lambda_2 = 2-i$$

$$\begin{pmatrix} 5 & -5 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{matrix} 0 \\ 1 \end{matrix}$$

(λ_1) $\begin{pmatrix} 5-2-i & -5 \\ 2 & -1-2-i \end{pmatrix}\begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ $\quad \begin{cases} 3u_1 - iu_1 - 5u_2 = 0 \\ 2u_1 - 3u_2 - iu_2 = 0 \end{cases}$ $\quad u_2(3-i) = 5u_2$

$$v_1 = \begin{pmatrix} 5 \\ 3 \end{pmatrix} + i\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$e^{\lambda_1 t} v_1 = e^{(2+i)t}\left(\begin{pmatrix} 5 \\ 3 \end{pmatrix} + i\begin{pmatrix} 0 \\ -1 \end{pmatrix}\right) = e^{2t}\left(\begin{pmatrix} 5 \\ 3 \end{pmatrix}\cos(t) - \begin{pmatrix} 0 \\ -1 \end{pmatrix}\sin(t) + i\begin{pmatrix} 5 \\ 3 \end{pmatrix}\sin(t) + i\begin{pmatrix} 0 \\ -1 \end{pmatrix}\cos(t)\right)$$

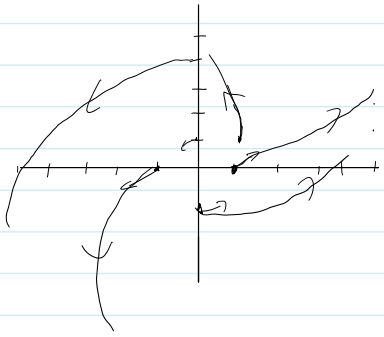
$$X_1(t) = e^{2t}\left(\begin{pmatrix} 5 \\ 3 \end{pmatrix}\cos(t) - \begin{pmatrix} 0 \\ -1 \end{pmatrix}\sin(t)\right)$$

$$x_1(t) = e^{2t}\left(\begin{pmatrix}5\\3\end{pmatrix}\cos(t) - \begin{pmatrix}0\\-1\end{pmatrix}\sin(t)\right)$$

$$x_2(t) = e^{2t}\left(\begin{pmatrix}5\\3\end{pmatrix}\sin(t) + \begin{pmatrix}0\\-1\end{pmatrix}\cos(t)\right)$$

$$X(t) = c_1 e^{2t}\begin{pmatrix}5\cos(t)\\3\cos(t)+\sin(t)\end{pmatrix} + c_2 e^{2t}\begin{pmatrix}5\sin(t)\\3\sin(t)-\cos(t)\end{pmatrix}$$

$$\begin{vmatrix}3 & 1\\ 2 & 0\end{vmatrix} = 0-2 = -2 \neq 0 \Rightarrow \text{linear independence}$$



1.5) Let V be an eigenvector of A, so $\exists \lambda$ s.t.

$$AV = \lambda V \quad \text{add } A^{-1} \text{ to both sides}$$
$$A^{-1}AV = \lambda A^{-1}V$$
$$1 \cdot V = \lambda A^{-1}V$$
$$\lambda^{-1}V = A^{-1}V$$

so for any V eigenvector of A with λ eigenvalue, V is also an eigenvector for A^{-1} with λ^{-1} as its eigenvalue.

1.6) Let's check if $au + bv + cw = 0$ only when $a=b=c=0$

$$a\begin{pmatrix}1\\3\\2\\1\end{pmatrix} + b\begin{pmatrix}4\\9\\5\\4\end{pmatrix} + c\begin{pmatrix}3\\7\\4\\3\end{pmatrix} = \begin{pmatrix}0\\0\\0\\0\end{pmatrix} \qquad \begin{cases} a+4b+3c=0 \\ 3a+9b+7c=0 \\ 2a+5b+4c=0 \\ a+4b+3c=0 \end{cases}$$

$a = -4b-3c$

$$-12b-9c+9b+7c=0$$
$$2c=-3b$$

$$-8b-6c+5b-6b=0$$
$$-9b+9b=0$$

so b is a free variable and it can be any non-zero number

So there are infinite solutions $\Rightarrow au+bv+cw=0 \Rightarrow u,v,w$ are linearly dependant

1.7)

$$A = \begin{pmatrix}1 & 0 & 2 & 2\\ 0 & 1 & -1 & -1\\ 2 & 4 & 0 & -1\end{pmatrix}$$

$$A=\begin{pmatrix}0&1&-1&-1\\2&4&0&-1\\-1&-2&1&2\end{pmatrix}$$

$$\left|\begin{pmatrix}1-\lambda&0&2&2\\0&1-\lambda&-1&-1\\2&4&-\lambda&-1\\-1&-2&1&2-\lambda\end{pmatrix}\right| = (1-\lambda)\begin{vmatrix}1-\lambda&-1&-1\\4&-\lambda&-1\\-2&1&2-\lambda\end{vmatrix} - 0 + 2\begin{vmatrix}0&1-\lambda&-1\\2&4&-1\\-1&-2&2-\lambda\end{vmatrix} - 2\begin{vmatrix}0&1-\lambda&-1\\2&4&-\lambda\\-1&-2&1\end{vmatrix}$$

lets find the eigenvalues -

$$(1-\lambda)\left[(1-\lambda)\left[(-\lambda)(2-\lambda)-(-1)(1)\right]-\left[(-1)\left[(4)(2-\lambda)-(4)(-2)\right]+(-1)\left[4(1)-(-2)(-\lambda)\right]\right]$$

$$+2\left[-(1-\lambda)\left[2(2-\lambda)-(-1)(-1)\right]+(-1)\left[2(-2)-(-1)(4)\right]\right]$$

$$-2\left[-(1-\lambda)\left[2(1)-(-\lambda)(-1)\right]+(-1)\left[2(-2)-(4)(-1)\right]\right] =$$

$$(1-\lambda)\left[(1-\lambda)(-2\lambda+\lambda^2+1)-(-1)(8-4\lambda-2)+(-1)(4-2\lambda)\right]$$

$$+2\left[-(1-\lambda)(4-2\lambda-1)+(-1)(-4+4)\right]$$

$$-2\left[-(1-\lambda)(2-\lambda)+(-1)(-4+4)\right]$$

$$\left[-2(1-\lambda)\right]\left[4-2\lambda-1-2+\lambda\right]$$

$$\left[-2(1-\lambda)\right]\left[1-\lambda\right]$$

$$(1-\lambda)\left[(1-\lambda)\underbrace{(-2\lambda+\lambda^2+1)}_{(1-\lambda)^2}-(-8+4\lambda+2)+(-4+2\lambda)\right]-2(1-\lambda)^2 =$$

$$(1-\lambda)\left[(1-\lambda)^3+8-4\lambda-2-4+2\lambda\right]-2(1-\lambda)^2 =$$

$$(1-\lambda)\left[(1-\lambda)^3+2-2\lambda\right]-2(1-\lambda)^2 = (1-\lambda)^4+\cancel{2(1-\lambda)^2}-\cancel{2(1-\lambda)^2} = (1-\lambda)^4$$

$$\det(A-\lambda I) = (1-\lambda)^4 = 0$$

$$1-\lambda = 0$$

$$\lambda = 1$$

lets find the eigenvectors

$$(A-\lambda I)V = 0$$

$$\begin{pmatrix}0&0&2&2\\0&0&-1&-1\\2&4&-1&-1\\-1&-2&1&1\end{pmatrix}\begin{pmatrix}u_1\\u_2\\u_3\\u_4\end{pmatrix} = 0$$

$$\begin{cases}2u_3+2u_4=0\\-u_3-u_4=0\\2u_1+4u_2-u_3-u_4=0\\-u_1-2u_2+u_3+u_4=0\end{cases} \Rightarrow \quad u_3=-u_4 \Rightarrow \quad 2u_1+4u_2-u_3+u_3=0$$

$$-u_1-2u_2+u_3-u_3=0$$

$\begin{pmatrix} \cdot & \cdot & -1 & -1 \\ -1 & -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} \cdot \\ u_4 \end{pmatrix}$ $\begin{cases} 2u_1 + u_2 - u_3 - u_4 = 0 \\ -u_1 - 2u_2 + u_3 + u_4 = 0 \end{cases} \Rightarrow$ $-u_1 - 2u_2 + u_3 - u_3 = 0$

$u_1 = 2u_2$

u_2 and u_4 are free variables, pick $u_2 = a$, $u_4 = b$

$v = \begin{pmatrix} -2a \\ a \\ -b \\ b \end{pmatrix}$ $\forall a, b$